

Eyenet-vision Platform

Version 2.23.1 | January 15, 2026

Core Functionalities & Features

1. Computer Vision & People Analytics

People Tracking & Analysis - Real-time people tracking and identification - Accurate counting and attendance monitoring - 3D route tracking and movement analysis - Demographic analysis (age, gender, etc.)

2. Camera & Hardware Integration

Multi-Source Camera Support - Traditional IP/network cameras - WiFi-enabled cameras - Soft cameras via mobile applications - Comprehensive camera health monitoring

3. Intelligent Triggers & Automated Actions

Event-Driven Automation - Camera-Level Trigger Configuration: Each camera can be independently configured with custom triggers based on machine vision events - **Real-Time Event Detection:** Automatic detection of configurable events (person count thresholds, demographics, dwell time, velocity, direction, etc.) - **Complex Condition Logic:** Support for multiple conditions with AND/OR operators for sophisticated trigger rules

Automated Action Execution - Multi-Channel Notifications - Mobile push notifications - In-app alerts - Email notifications - SMS alerts (via integrations) - **Digital Signage Integration:** Automatic content switching on connected displays based on audience demographics or behaviors - **Third-Party Application Triggers:** Webhook-based actions to external systems (access control, lighting, HVAC, etc.) - **Custom Workflows:** Programmable action chains for complex automation scenarios

Use Case Examples - Retail: Trigger digital signage content change when high-value demographic detected - Security: Alert security personnel when person count exceeds threshold in restricted area - Smart Buildings: Adjust lighting/HVAC based on occupancy and movement patterns - Queue Management: Send alerts when queue length exceeds acceptable wait time - Access Control: Trigger door locks or alerts based on unauthorized presence detection

4. Video & Data Management

Storage & Processing - Local video storage and management - No cloud services required for video/data storage - Full operational capability without internet connectivity - Minimal internet access needed only for remote notifications and email

5. Intelligent Communications

Multi-Channel Notification System - Mobile push notifications - In-app alerts and warnings - Email notifications - Automatic integrations with third-party applications - Webhook support for custom workflows - Audit logging of all communications with delivery tracking

6. Analytics & Insights

Data Intelligence - Real-time analytics dashboards - Historical trend analysis - Behavioral pattern recognition - Actionable insights and reporting - Trigger performance analytics - Action execution monitoring and statistics

7. Deployment Architecture

Local & Edge Installation - Complete on-premises deployment - No dependency on cloud services - Edge computing architecture for data privacy - Internet connectivity optional (required only for remote features)

Microservices Design - Modular microservices architecture with separation of concerns - Independent service scaling and maintenance - Automatic service discovery and health monitoring - Environment-aware installations for remote operation tracking

8. Security & Access Control

Enterprise-Grade Security - Comprehensive user access management with role-based permissions - File-level encryption for sensitive data - Optional private VPN for internet communications - Complete data sovereignty (all data stays on-premises) - Secure trigger and action configuration with user audit trails

9. Scalability & Expandability

Flexible Scaling - Scales from small installations (homes) to large industrial sites - Hardware-based scalability (software adapts automatically) - Performance scales with available hardware resources - No architectural limits on deployment size - Unlimited camera support based on hardware capacity

Integration & Extensibility - Third-party software connectivity via REST APIs - Standard webhook integrations for custom actions - Custom integration support - Modular plugin architecture - Digital signage system integrations (standalone and networked displays)

10. Support & Monitoring

Remote Support Capabilities - Remote diagnostic and troubleshooting tools - Centralized logging and monitoring - Installation identification for multi-tenant support - Comprehensive audit trails for compliance - Trigger execution monitoring and debugging - Action delivery tracking and failure analysis

Key Differentiators

- ❑ **Privacy-First:** All processing and storage on-premises
- ❑ **Offline-Capable:** Full functionality without internet
- ❑ **Infinitely Scalable:** From single camera to enterprise-wide
- ❑ **Secure by Design:** Encrypted, access-controlled, VPN-ready
- ❑ **Integration-Ready:** REST APIs, webhooks, third-party connectors
- ❑ **Support-Friendly:** Remote diagnostics without compromising security
- ❑ **Event-Driven Intelligence:** Automated actions based on real-time vision analytics
- ❑ **Per-Camera Automation:** Independent trigger configuration for each camera
- ❑ **Multi-Action Orchestration:** Complex workflows with multiple notification channels and integrations

Technical Architecture Highlights

Trigger & Action System

- **Granular Configuration:** Each camera operates independently with its own trigger rules
- **Real-Time Processing:** Instant evaluation of machine vision events against trigger conditions
- **Reliable Delivery:** Communication service ensures action execution with retry logic and failure tracking

- **Comprehensive Logging:** Full audit trail of trigger activations and action executions with multi-tenant support

Operational Environment Awareness

- **Context-Aware Triggers:** Conditions based on time of day, day of week, occupancy levels, demographics
- **Behavioral Pattern Recognition:** Triggers based on velocity, direction, dwell time, and trajectory
- **Historical Comparison:** Trigger conditions comparing current vs. historical averages
- **Multi-Camera Correlation:** Cross-camera triggers for tracking across multiple zones

Digital Signage Integration

- **Dynamic Content Switching:** Automatic display content changes based on audience detection
- **Demographic-Based Targeting:** Show relevant content based on detected age, gender demographics
- **Occupancy-Aware Display:** Adjust content based on crowd size and engagement metrics
- **Playlist Automation:** Intelligent scheduling based on real-time analytics

This platform provides a complete vision analytics solution with intelligent automation, enabling businesses to transform passive surveillance into active, responsive operational intelligence.